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Precise and non-destructive approach for identifying the real concentration based on cured cemented paste backfill using hyperspectral imaging

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Abstract: Cemented paste backfill (CPB) is a technology that achieves safe mining by filling the goaf with waste rocks, tailings, and other materials. It is an inevitable choice to deal with the development of deep and highly difficult mines and meet the requirements of environmental protection and safety regulations. It promotes the development of a circular economy in mines through the development of low-grade resources and the resource utilization of waste, and extends the service life of mines. The mass concentration of solid content (abbreviated as "concentration") is a critical parameter for CPB. However, discrepancies often arise between the on-site measurements and the pre-designed values due to factors such as groundwater inflow and segregation within the goaf, which cannot be evaluated after the solidification of CPB. This paper innovatively provides an *in-situ* non-destructive approach to identify the real concentration of CPB after curing for certain days using hyperspectral imaging (HSI) technology. Initially, the spectral variation patterns under different concentration conditions were investigated through hyperspectral scanning experiments on CPB samples. The results demonstrate that as the CPB concentration increases from 61wt% to 73wt%, the overall spectral reflectance gradually increases, with two distinct absorption peaks observed at 1407 and 1917 nm. Notably, the reflectance at 1407 nm exhibited a strong linear relationship with the concentration. Subsequently, the K-nearest neighbors (KNN) and support vector machine (SVM) algorithms were employed to classify and identify different concentrations. The study revealed that, with the KNN algorithm, the highest accuracy was achieved when K (number of nearest neighbors) was 1, although this resulted in overfitting. When $K = 3$, the model displayed the optimal balance between accuracy and stability, with an accuracy of 95.03%. In the SVM algorithm, the highest accuracy of 98.24% was attained with parameters C (regularization parameter) = 200 and Gamma (kernel coefficient) = 10. A comparative analysis of precision, accuracy, and recall further highlighted that the SVM provided superior stability and precision for identifying CPB concentration. Thus, HSI technology offers an effective solution for the *in-situ*, non-destructive monitoring of CPB concentration, presenting a promising approach for optimizing and controlling CPB characteristic parameters.

Keywords: cemented paste backfill; concentration; hyperspectral imaging; non-destructive testing

1. Introduction

Mineral resources serve as a crucial foundation for national economic development and technological advancement [1–4]. However, large-scale resource extraction often results in the significant discharge of solid wastes [5–8], including tailings, phosphogypsum, and coal gangue [9]. These wastes not only occupy extensive land areas but also contribute to severe pollution of surrounding land and groundwater [10–13]. Cemented paste backfill (CPB) offers an effective solution to these challenges by mixing solid waste, cement, and water to create a backfill slurry, which is subsequently transported to the goaf. This method addresses the issues of solid waste disposal and the potential hazards posed by void spaces in mined-out areas [14–16].

The mass concentration of CPB is a critical factor affecting its performance. CPB with higher concentrations generally exhibits greater strength and a more compact microstructure [3,17–18]. The concentration of CPB can be effectively controlled and monitored during its preparation. However, once the CPB reaches the goaf, factors such as groundwater inflow or segregation often lead to discrepancies between the actual concentration and the intended concentration [19–20]. This deviation is a key reason for the variation in CPB performance from the design specifications. Once the CPB has cured for a certain period or when dealing with unfamiliar samples, determining the concentration via visual inspection or basic equipment becomes challenging. In mining operations, for instance, CPB quality is typically monitored using core sampling, followed by uniaxial compressive strength tests to evaluate its strength. However, the CPB concentration remains unknown, and it is uncertain whether the observed strength differences are due to variations in concentration [21]. It is well-known that the con-

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centration measurement of fresh CPB slurry is very easy, while there are no effective methods for identifying the real concentration of CPB after curing for certain d in the goaf. Therefore, under the conditions encountered in the goaf, the ability to rapidly and non-destructively identify the actual concentration of CPB is crucial for accurately controlling backfill parameters and facilitating intelligent backfilling. Furthermore, the absence of real-time non-destructive concentration testing methods for cured CPB renders backfilling strategies difficult to adjust in a timely manner, potentially leading to risks of insufficient filling quality or goaf instability. Secondly, traditional core sampling is not only time-consuming and labor-intensive but also destructive, rendering it incapable of enabling large-scale continuous monitoring of CPB in goafs. Thus, there remains an urgent need for a rapid, non-destructive, and precise detection method for CPB concentration.

Hyperspectral imaging (HSI) is a powerful technology that combines both spectral and imaging capabilities [22–23]. It enables for the collection of detailed information about an object's composition and characteristics unattainable by traditional imaging systems [24]. The reflectance curve represents the interaction between electromagnetic waves and objects. A key feature of HSI is its integration of spectra and images, which has led to its growing popularity in various industries and research fields due to its non-invasive and non-destructive capabilities for material identification and quantification [25–27]. When it comes to concentration, HSI demonstrates superior performance. For instance, Guo *et al.* [28] studied chromium (Cr) concentration and proposed an optimal model for estimating heavy metal concentrations in soil. This model, which incorporated spectral preprocessing, optimal parameter dimensionality reduction, and various models, achieved a residual prediction deviation of 2.04. Achata *et al.* [29] developed a partial least squares (PLS) regression model to identify and predict brining salt concentrations in pork (with salting concentrations of 5wt%, 10wt%, and 15wt%), achieving a correlation coefficient of 0.92. Given HSI's high sensitivity to water, it has also been used to detect moisture content in cement systems. Gastaldi *et al.* [30] analyzed the moisture content and ettringite (AFt) formation during the hydration of calcium sulfoaluminate cement using HSI. They found that as the calcium aluminate content increased, the absorption peak areas at wavelengths of 7000 and 5250 cm^{-1} progressively increased. Similarly, Gastaldi *et al.* [31] employed near-infrared spectroscopy to study the degree of cement hydration, observing that as the hydration process advanced, the amount of hydration products increased and their characteristic peaks became more pronounced. These findings were consistent with hydration degree calculations.

HSI is being increasingly applied across various fields, yet its application in the backfill domain remains relatively under explored. By acquiring spectral data of CPB with different mix proportions, this study analyzed the spectral response patterns under varying concentrations and curing ages, establishing the correlation between spectral characteristics and CPB concentration. Based on HSI data, K-nearest neighbors (KNN) and support vector machine (SVM) concentration identification models were constructed and compared to explore the feasibility of HSI for cured CPB concentration identification. This research provides a theoretical basis for the *in-situ* non-destructive testing of CPB concentration in stopes, thereby enabling precise control of backfilling parameters and intelligent backfilling operations.

2. Experimental

2.1. Materials

The experimental materials used in this study include tailings, cement, and water. The tailings were sourced from the Yinshan Mine in Jiangxi Province, China, and the cement used was ordinary Portland cement 42.5 (P.O 42.5). The chemical compositions of both the tailings and cement were analyzed using X-ray fluorescence spectrometry (XRF) [32], with the results presented in Table 1. As shown in Table 1, the primary chemical components of the tailings are SiO_2, Al_2O_3, and Fe_2O_3, while the cement contains the highest proportion of CaO. Additionally, a laser particle size analyzer was employed to analyze the particle size distribution of the tailings. Fig. 1 presents the particle size distribution, indicating that the tailings used in this experiment are classified as fine tailings.

2.2. CPB preparation

Different concentrations of CPB were prepared by varying the mass ratios of water to solids (the total of cement and tailings). Based on mining production practices and to achieve higher strength, the cement-to-tailings mass ratio was set at 1:4, with mass concentration gradients of 61wt%, 64wt%, 67wt%, 70wt%, and 73wt%. The tailings, cement, and water were weighed according to the specified mix proportions. The materials were mixed and poured into a stirring bucket, then agitated with a handheld mixer for 5 min. Subsequently, the backfill slurry was transferred into standard triple molds (7.07 cm × 7.07 cm × 7.07 cm). During casting, a glass rod was used to vibrate the slurry for thorough air bubble removal. After 1.5 d, excess backfill on the mold surface was scraped off, and the specimens were demolded, immediately placed in a standard curing chamber at 25°C with 90% humidity. For each concentration gradient, nine specimens were prepared. At curing ages of 3, 7, 11, 14, 17, 21,

Table 1. Chemical compositions of experimental materials wt%

Material	SiO_2	Al_2O_3	Fe_2O_3	SO_3	K_2O	MgO	TiO_2	CaO	P_2O_5	Na_2O
Tailings	59.63	20.88	6.837	4.795	4.722	1.13	0.716	0.589	0.151	0.15
Cement	21.81	5.02	3.95	3.72	0.79	2.22	0.32	58.46	0.12	0.10

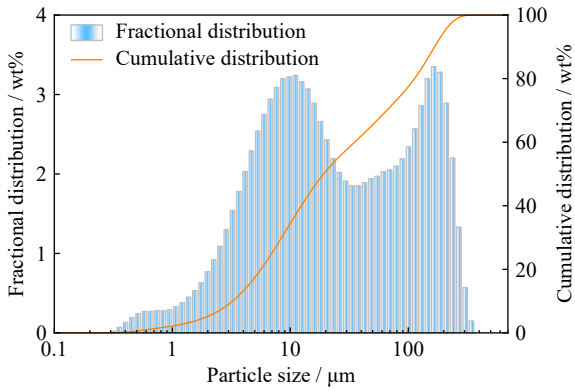

Fig. 1. Particle size distribution curve of tailings.

25, and 28 d, one specimen per concentration was extracted for hyperspectral scanning, as illustrated in Fig. 2.

To investigate spectral responses of CPB in different morphologies, hyperspectral scanning was conducted on four lateral surfaces of specimens cured for 7 d. Note that as the bottom surface was labeled, the scanning targets excluded the top and bottom, designating the four lateral surfaces as Side-1 to Side-4. The specimens were then fractured along the central plane to expose the internal cross-section, which underwent hyperspectral scanning. Finally, the specimens were crushed into fragments, and spectral testing was repeated following the same procedure.

2.3. Hyperspectral scanning

The surfaces of the CPB samples were scanned using the GaiaField-DZ portable spectrometer, manufactured by Jiangsu Shuanglihe Spectrum Technology Co., Ltd. This device is equipped with a hyperspectral camera that covers the wavelength range of 1000 to 2500 nm (shortwave near-infrared), with a spectral resolution of 5.6 nm and a total of 288 bands. The entire instrument was housed inside a sealed black box to prevent external light from influencing the experiment. Four halogen lamps were installed within the black box to serve as the light source. The Specview software was used to control the spectrometer, including adjustments for the camera's focal length and exposure settings. Prior to testing, a 100% reflectance white board was used for calibration, assuming zero absorption across all bands. To ensure accurate imaging, the CPB samples were positioned directly beneath the hyperspectral camera at the center of the image, and each sample was scanned sequentially according to its identification number.

2.3.1. Spectral data acquisition

The scanned files were imported into Specview software for per-pixel reflectance correction, with the calculation formula provided in Eq. (1). Simultaneously, spectral smoothing was applied to mitigate the effects of noise and spectral spikes. The corrected spectra were then extracted using ENVI 5.6 software, with the entire surface of the CPB samples serving as the focus of the analysis. A 5 pixel × 5 pixel block was selected as the region of interest (ROI). Each individual ROI was treated as a whole, and the average spectrum for that region was calculated. To minimize errors, approximately 200 ROIs were extracted from each CPB sample, and the average spectrum of these 200 ROIs was used as the hyperspectral data for the samples.

$$R_{\mathrm{ref}} = \frac{\mathrm{DN_{raw}} - \mathrm{DN_d}}{\mathrm{DN_w} - \mathrm{DN_d}} \tag{1}$$

where R_{ref} is the corrected reflectance spectrum, $\mathrm{DN_{raw}}$ is the

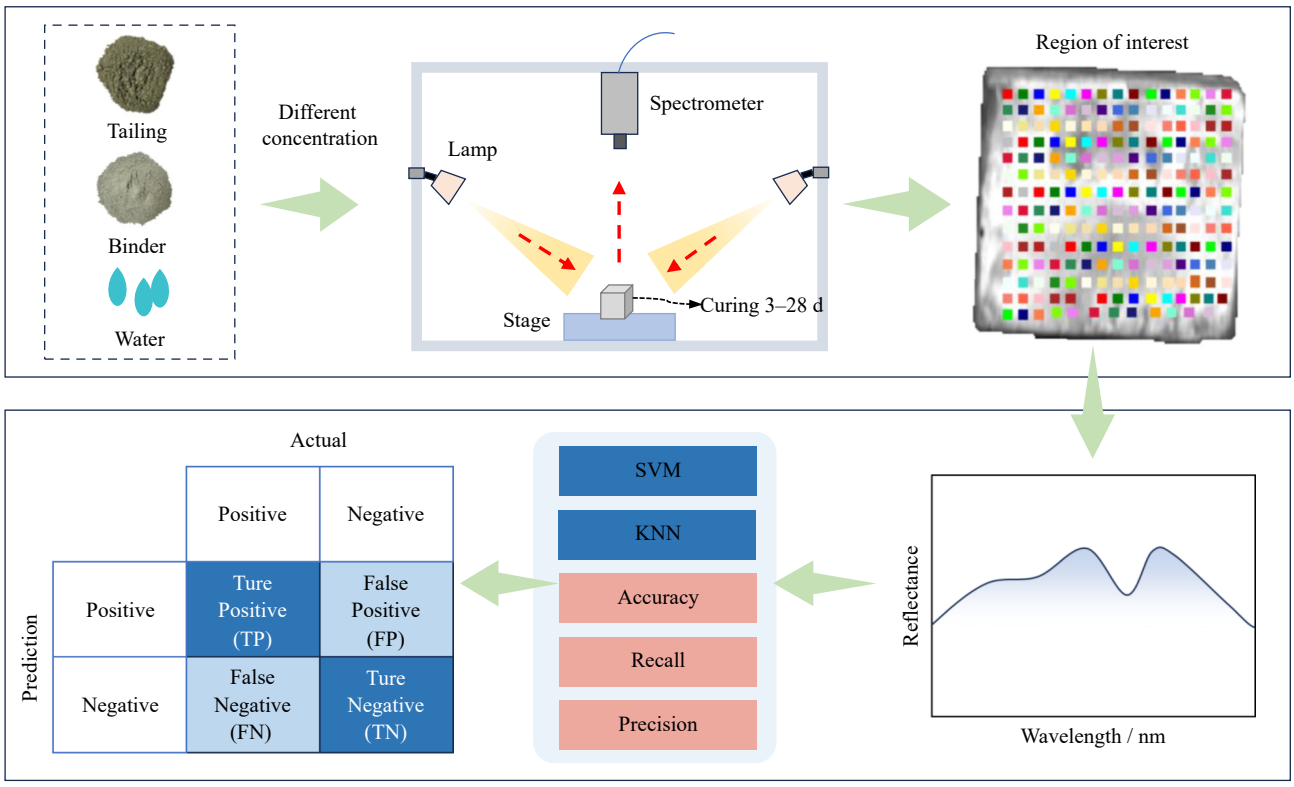

Fig. 2. Hyperspectral scanning and CPB concentration identification process.

unprocessed raw reflectance spectrum, DN_d is the reflectance spectrum under dark background conditions, and DN_w is the reflectance spectrum under white reference background conditions.

2.3.2. Spectral pretreatment

Due to background interference, the irregular shape of the spectrum complicates feature comparison and identification. To address this, spectral feature enhancement can be applied by removing specific noise or redundant information, thereby extracting a cleaner spectral signal [33]. Continuous removal (CR) of spectral noise is a crucial step in this process. Firstly, local maximum points are identified, which are considered to represent the "peak" positions of data in the region and serve as key control points for constructing the envelope. After determining the peak points, cubic spline interpolation is employed to fit these points. This method ensures that the generated envelope is tangent to the original data at each control point and remains smooth and continuous across the entire data interval, thereby yielding the final envelope. The calculation formula is shown in Eq. (2), and the schematic diagram is illustrated in Fig. 3.

$$S_{CR} = \frac{R_S}{R_{CR}} \tag{2}$$

where S_{CR} is continuum removed spectrum, R_S is original spectrum, and R_{CR} is continuum curve.

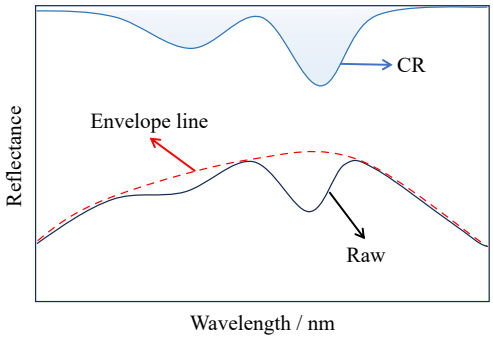

Fig. 3. Diagram of the spectral CR process.

3. Machine learning algorithm model

Several algorithms are commonly used for classification, including SVM, KNN, random forest, gradient boosting machine, and others [34–35]. SVM is particularly effective for handling high-dimensional data, especially in cases where the number of features exceeds the number of samples. In contrast, KNN classifies data based on the inherent characteristics of the data itself, making it highly adaptable. Due to these advantages, both SVM and KNN are widely applied in the classification of multi-dimensional hyperspectral data [36].

3.1. SVM

SVM is a robust supervised learning algorithm widely used for classification and regression tasks. The model can be constructed by relying solely on a small number of "support vectors", making it suitable for scenarios with limited hyperspectral data samples. Numerous scholars have demonstrated the superiority of the SVM algorithm in classification tasks [37–38].

In the context of multi-class problems with hyperspectral data, the objective of SVM is to select a hyperplane that maximizes the margin between classes, ensuring that the minimum distance between classes is as large as possible on that hyperplane. In a two-dimensional space, this hyperplane can be represented as a linear equation as Eq. (3) [39]:

$$\omega^{T} x + b = 0 \tag{3}$$

where $\boldsymbol{\omega}$ is the vector perpendicular to the hyperplane, which determines the direction of the hyperplane, b is the bias term, and x is the input data. For a given sample point x_i, with its class label y_i, the distance (d_i) to the hyperplane is calculated as Eq. (4):

$$d_i = \frac{\left|\omega^{T} x_i + b\right|}{\|\omega\|} \tag{4}$$

The optimization problem of SVM can be transformed into the following:

$$\min J = \frac{1}{2}\|\omega\|^2 \tag{5}$$

where J is the objective function.

In the case of more general, non-linearly separable data, SVM can transform the above problem into a minimization problem with a penalty term using soft margin optimization and the kernel trick [40]. The transformed problem is as Eq. (6):

$$\begin{cases} \min J = \frac{1}{2}\omega^{T} x_i + C\sum_{l}^{i=1}\xi_i \\ \text{s.t. } y_i\left[\omega^{T}\varphi(x_i) + b\right] \geq 1 - \xi_i \\ \xi_i \geq 0 \quad i = 1, 2, \cdots, l \end{cases} \tag{6}$$

In this case, b is the bias term, $C > 0$ is the penalty factor, ξ_i are the slack variables, and φ is the mapping function. The input data x_i is mapped into a higher-dimensional space by φ, and SVM finds a linear hyperplane in this high-dimensional space that maximizes the margin to classify the data. HSI typically encompasses hundreds to thousands of continuous spectral bands, carrying abundant information on material compositions. The relationship between CPB concentration and spectral features may involve complex nonlinear mappings, which can be addressed by transforming original hyperspectral data into a high-dimensional feature space through nonlinear mapping, thus resolving nonlinear classification problems. The kernel function can be defined as Eq. (7):

$$K(x_i, x_j) = \varphi^{T}(x_i)\varphi(x_j) \tag{7}$$

where K is the kernel function. In SVM, commonly used kernel functions include the polynomial kernel, Gaussian kernel, and Sigmoid kernel, as shown in Eq. (6). The choice of kernel function and its parameters currently lack a specific set of operational guidelines. In general, the Gaussian kernel is considered a reliable kernel, as it does not introduce significant deviations. Here are the commonly used kernel functions:

$$\text{polynomial}: K(x_i, x_j) = \left(\gamma x_i^{\mathrm{T}} x_j + r\right)^d \quad \gamma > 0$$
$$\text{RBF}: K(x_i, x_j) = \exp\left(-\gamma x_i - x_j^2\right) \quad \gamma > 0$$
$$\text{sigmoid}: K(x_i, x_j) = \tanh\left(\gamma x_i^{\mathrm{T}} x_j + r\right) \tag{8}$$

where r, d, and γ are parameters in the kernel function respectively.

3.2. KNN

The core principle of the KNN algorithm for hyperspectral classification is based on spectral vector similarity measurement. Classification is achieved by identifying the K nearest neighbor samples of an unknown sample within the training set and using their class labels for voting or weighted averaging. First, hyperspectral data is converted into multidimensional spectral vectors, and distance metrics—such as Euclidean distance, Mahalanobis distance, or spectral angle mapper (SAM)—are used to calculate inter-sample distances. For an unknown sample, K nearest neighbors are determined, and classification is completed through majority voting. The common Euclidean distance formula is as Eq. (9) [41]. Well-suited for small-sample scenarios with flexible sample updating, KNN does not require model retraining. The effectiveness of the KNN algorithm in processing hyperspectral data has been fully validated in the fields of food science and agriculture [42–43].

$$d(z_i, z_j) = \sqrt{\sum_{n}^{k=1} (z_{ik} - z_{jk})^2} \tag{9}$$

where $d(z_i, z_j)$ represents the distance between the z_i and z_j; z_{ik} and z_{jk} are the values of the k-th feature for the samples z_i and z_j, respectively; n is the number of dimensions of the sample.

3.3. Evaluation index

Common evaluation metrics in classification problems include Accuracy, Recall, Precision, and F1-Score [44]. In this section, the classification performance of two algorithms is compared using three evaluation metrics: Accuracy, Recall, and Precision, based on the experimental results for text data classification.

Precision measures the model's ability to correctly predict positive class samples. The formula for Precision is as Eq. (10).

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{10}$$

Recall measures the model's ability to identify as many positive class samples as possible. The formula for Recall is as Eq. (11).

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{11}$$

Accuracy is a commonly used metric to measure the overall performance of a classification model. It represents the proportion of correctly predicted samples out of the total number of samples. The formula for Accuracy is as Eq. (12).

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{12}$$

where TP is the positive score predicted as positive score, TN is the negative score predicted as negative score, FP is the negative score predicted as positive score, and FN is the positive score predicted as negative score.

4. Results and analysis

4.1. The spectral analysis of tailings

As shown in Fig. 4(a), the spectral curve of the tailings reveals three main absorption peaks at approximately 1390, 1950, and 2200 nm. It should be noted that due to the relatively high spectral noise in the 2400–2500 nm band, only the range before 2400 nm is analyzed. The peak around 1390 nm corresponds to the first overtone stretching vibration of the O–H bond in H_2O molecules ($2\nu_1$, $2\nu_3$, and $\nu_1 + \nu_3$) [30]. The absorption peak at 1950 nm arises from a combination of stretching and bending vibrations of the O–H bond. It is important to note that the two absorption peaks attributed to H_2O are relatively weak and broad, which can be attributed to the drying process applied to the tailings before spectral measurement, leading to a very low moisture content. Nevertheless, due to hyperspectral sensitivity to water, even small amounts of moisture can produce weak absorption peaks [45]. A strong absorption peak, which is higher than the O–H absorption, appears near 2200 nm. This absorption in the 2200 nm range is attributed to the combination of Si–O and Al–O bonds [46]. The presence of this absorption peak indicates that the tailings are rich in SiO_2 and Al_2O_3, which is consistent with the XRF analysis results mentioned earlier [47]. Fig. 4(b) illustrates the mineralogical phase composition of

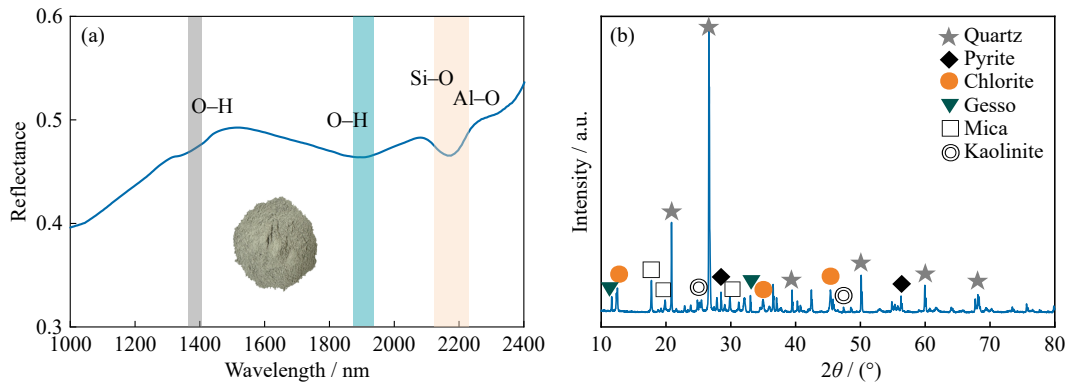

Fig. 4. The spectral curve and X-ray diffraction analysis of tailings: (a) hyperspectral; (b) X-ray diffraction analysis.

tailings. A strong diffraction peak at 2θ of 26.5° indicates the presence of a large number of quartz crystals, which can also support the above view. In addition, tailings contain other mineral phases such as pyrite, chlorite, and mica.

4.2. The spectral response characteristics of CPB

4.2.1. Different CPB morphology

Fig. 5 illustrates the spectral response characteristics of the CPB in various form. Fig. 5(a) shows that two distinct absorption peaks are observed approximately 1407 and 1917 nm. However, the overall reflectance baseline of the spectrum exhibits significant variations. The spectra of the four sides of the CPB display similar characteristics, while the fragmented pieces and cross-sections show comparable spectra. The reflectance values of the four sides of the CPB are higher, which is directly related to the roughness, particle size, and moisture content of the material being tested [48]. Compared to the cross-sections and fragments, the sides of the CPB are smoother, resulting in increased light reflection. Consequently, the sensor detects a stronger digital number (DN), leading to higher overall reflectance. Despite the differences in overall reflectance intensity, the positions of the two main absorption peaks at 1407 and 1917 nm remain consistent. This indicates that the spectra of the backfill in various states still retain key information and distinct "fingerprint" features.

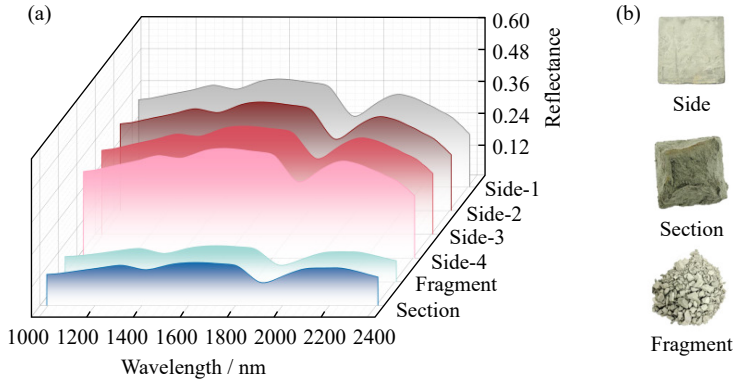

Fig. 5. Spectral characteristics under different CPB morphology: (a) spectral reflectance and (b) different states.

4.2.2. The spectral variation patterns of different concentrations

Fig. 6 shows the spectral response curves of the CPB under different concentration conditions. As illustrated in Fig. 6 (a), the reflectance gradually increases with higher CPB concentration, with the highest reflectance observed at a concentration of 73wt%. This overall increase in reflectance is associated with the moisture content of the CPB. Under identical conditions, a higher CPB concentration results in lower free water content, leading to less absorption of incident light [49]. Within the 1000 to 2400 nm wavelength range, three absorption peaks appear at 1407, 1917, and 2178 nm. The positions of these peaks remain consistent and are unaffected by the CPB concentration. Among these peaks, the absorption peak at 1917 nm is the strongest, followed by the 1407 nm peak. The absorption peak at 2178 nm is weak and difficult to resolve in the raw spectral curve. To enhance the absorption features for detailed observation and analysis, the original spectra were preprocessed using the CR method. The processed spectra are shown in Fig. 6(b). As seen in Fig. 6(b), with increasing CPB concentration, the area of the absorption peaks at 1407 and 1917 nm gradually decreases. Based on the earlier analysis, these absorption peaks at 1407 and 1917 nm are attributed to the stretching and bending vibrations of the O–H bond in H_2O molecules. The higher the O–H content, the greater the energy required for molecular-level transitions, resulting in a stronger absorption peak.

However, at the 2178 nm wavelength, an opposite trend is observed, with the absorption peak area increasing as the CPB concentration rises. At a concentration of 61wt%, there is virtually no absorption peak at this wavelength. As the concentration increases to 73wt%, a narrow and deep absorption peak emerges, which is more pronounced compared to the absorption peaks at other concentrations. Based on previous studies, this wavelength is associated with the vibration of AFt, Si–O, and Al–O bonds [30]. The CPB is a complex hydration system that contains both raw materials and hydration products. In ordinary Portland cement hydration systems, hydration products typically result from the reactions between $3CaO{\cdot}Al_2O_3$ (C_3A), $3CaO{\cdot}SiO_2$ (C_3S), and water. When C_3A and C_3S come into contact with water, a significant amount of Ca^{2+} and silicate ions are released during the hydration process [50]. As the concentration of Ca^{2+} and silicate ions increases in the solution, precipitates such as calcium silicate hydrate (C–S–H), $Ca(OH)_2$, and AFt are formed [51–52]. These precipitates are the primary contributors to strength development and the formation of dense structures. The basic structural unit of C–S–H consists of a layered structure formed by silicon–oxygen tetrahedra (SiO_4) connected through shared corner oxygen atoms [53]. The amount of C–S–H generated is directly related to the cement clinker content and curing time. Compared to other concentrations, the 73wt% concentration of CPB contains more cement clinker and corresponding hydration products (Si–O bonds), a conclusion supported by several studies. Therefore, it is hypothesized that the absorption peak at 2178 nm is due to the formation of the hydration product C–S–H. With the decrease in backfill concentration, the depth of absorption peaks gradually diminishes, and when the concentration reaches 61wt%, the absorption peaks are barely observable.

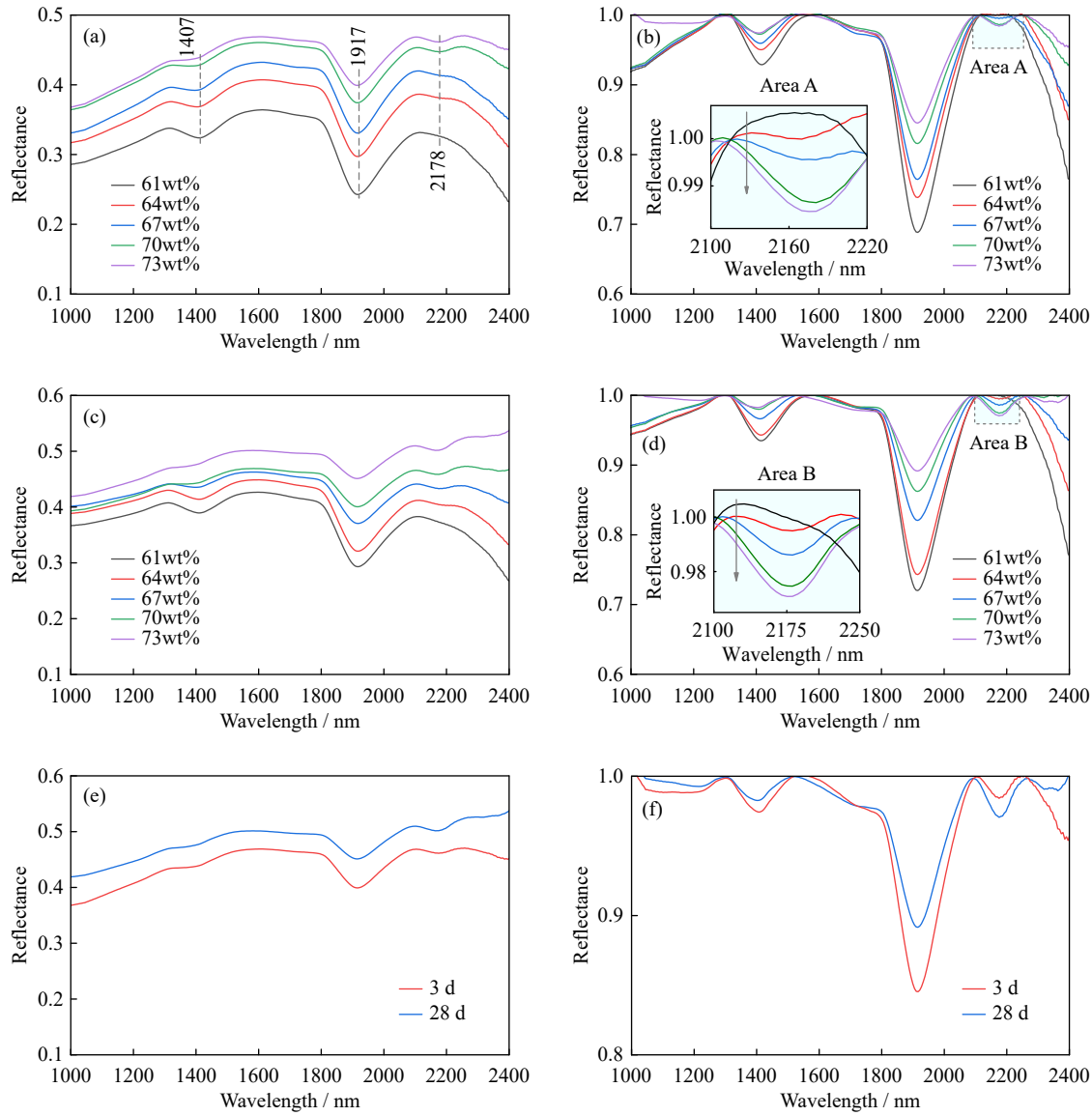

Fig. 6. Spectral variation patterns of CPB at different concentrations and curing ages: raw (a) and CR-processed (b) spectra at 3 d curing age; raw (c) and CR-processed (d) spectra at 28 d curing age; raw (e) and CR-processed (f) spectra at 73wt% concentration.

This is because a lower concentration corresponds to a reduced cement content, leading to fewer hydration products such as C–S–H and $Ca(OH)_2$ produced under identical conditions. According to the Lambert-Beer law, the lower the substance content, the weaker the absorption capacity for light at specific bands, causing the spectral curve to exhibit flatter absorption peaks.

It is important to further clarify that both the tailings and cement raw materials are rich in SiO_2. Whether the changes in the absorption peak at the 2178 nm wavelength are due to the increase in solid content requires further analysis. To test this hypothesis, spectral curves of the CPB under 73wt% concentration but different curing times were plotted, as shown in the Fig. 6(e) and (f). At a curing age of 3 d, the absorption peak intensity at 2178 nm is smaller than at 28 d. Therefore, it can be concluded that the absorption peak at 2178 nm is not caused by the increased solid content of SiO_2, thus ruling out this factor. This further supports the conclusion that the absorption peak at 2178 nm is attributed to the hydration product C–S–H. As hydration time increases, more hydration products are formed. Additionally, at a curing age of 3 d, the absorption intensities at 1407 and 1917 nm are stronger. This phenomenon can be explained by the fact that the hydration reaction and evaporation processes have consumed a certain amount of free water, resulting in reduced absorption in the "water-sensitive" wavelength bands of the incident light. Thus, the stronger absorption at 1407 and 1917 nm at 3 d is likely due to the presence of more free water before the hydration process is completed.

Based on the analysis above, the reflectance values at the 1407 and 1917 nm wavelengths are particularly sensitive to changes in concentration. Since the 1917 nm absorption peak corresponds to the combination of stretching and bending vibrations of the Al–O bond and H_2O molecules, the reflectance at 1407 nm was selected for fitting with the CPB concentration, as shown in Fig. 7. As seen in Fig. 7, the reflectance at 1407 nm shows a linear relationship with concentration over curing ages from 3 to 28 d, with an R^2 value con-

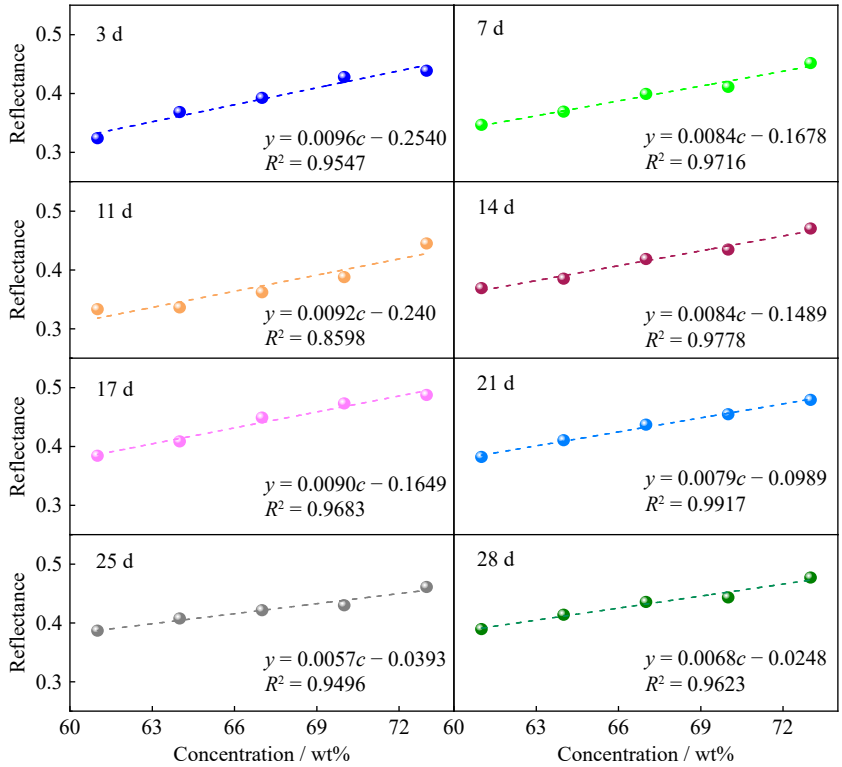

Fig. 7. Relation between CPB concentration and reflectance (At 1407 nm).

sistently exceeding 0.9, indicating a high level of reliability. This suggests that the reflectance at 1407 nm can be effectively used to quantify the CPB concentration.

4.3. Concentration identification

The above study demonstrated a strong linear relationship between the CPB concentration and the reflectance at the 1407 nm, indicating that hyperspectral reflectance offers valuable information for identifying CPB concentration. The next step is to apply machine learning techniques to recognize both the CPB concentration and the curing time.

4.3.1. Hyperparameter optimization

To improve the model's generalizability and robustness, the Kennard-Stone algorithm is employed to divide the data into 70% for the training set and 30% for the testing set. This approach ensures that the model is trained on a substantial portion of the data while also evaluating its performance on a separate set of unseen data. This provides a better estimate of the model's true accuracy and reduces the risk of overfitting.

Both KNN and SVM algorithms are employed to develop classification models using the training data, followed by prediction of the classification labels for the test data. In the SVM algorithm, tuning hyperparameters is essential for optimizing the model's performance. The primary hyperparameters include C (regularization parameter) and Gamma (kernel coefficient) [54]. C regulates the model's tolerance for misclassifications in the training data, while Gamma determines the influence range of each training sample, thereby affecting the decision boundary of the model. In this study, values for C were selected from the range (0.1, 1.0, 10, 100, 200, 500), and values for Gamma were chosen from (0.001, 0.01, 0.1, 1, 10). A grid search optimization algorithm, combined with 5-fold cross-validation, was used to optimize these hyperparameters, with the optimization results presented in the Fig. 8. For classification based on curing time, the highest accuracy of 99.35% was achieved with $C = 500$ and Gamma = 1. For classification based on concentration, the highest accuracy of 93.27% was achieved with $C = 200$ and Gamma = 10. For classification based on sample blocks, the highest accuracy of 96.29% was achieved with $C = 500$ and Gamma = 1. These results demonstrate the effectiveness of using SVM with optimized hyperparameters for classifying different CPB properties.

In the KNN algorithm, the value of K (number of nearest neighbors) was varied from 1 to 30, and a loop was employed to analyze the prediction accuracy of the algorithm for different K values. Fig. 8(d) shows the classification accuracy on the training set for various K values. When $K = 1$, the accuracy is highest, while $K = 3$ provides the second-highest accuracy. When K is too small, the KNN classifier may become overly sensitive to noise points in the data, leading to overfitting. This is reflected in the model's excellent performance on the training set but poor performance on the test set [55]. To mitigate the risk of overfitting with $K = 1$, both $K = 1$ and $K = 3$ were selected for further analysis. By comparing these two values, the impact of overfitting can be evaluated, allowing for more robust predictions.

4.3.2. Identification result

As shown in Table 2, the SVM classification results for the CPB indicate that, regardless of the classification metric, the accuracy can reach 98.50%. Notably, the classification accuracy for the curing age of the CPB is the highest, reaching 99.84%. Additionally, the accuracies for the training and test sets are quite close, with no overfitting observed. In the

KNN classification model, when $K = 1$, the training set accuracy reaches 99.9%, but the test set accuracy is only 94.2%, indicating significant overfitting. As a result, further analysis of this scenario is not warranted. As shown in Table 3, the KNN classification results for the CPB with $K = 3$ yield the following classification accuracies for the test set: 92.64% for blocks, 95.03% for concentration, and 95.26% for curing time. These results demonstrate improved performance compared to $K = 1$ and indicate that $K = 3$ provides more balanced and reliable predictions for all categories.

Fig. 9 shows a comparison of the classification accuracy between the SVM and KNN algorithms. It can be observed

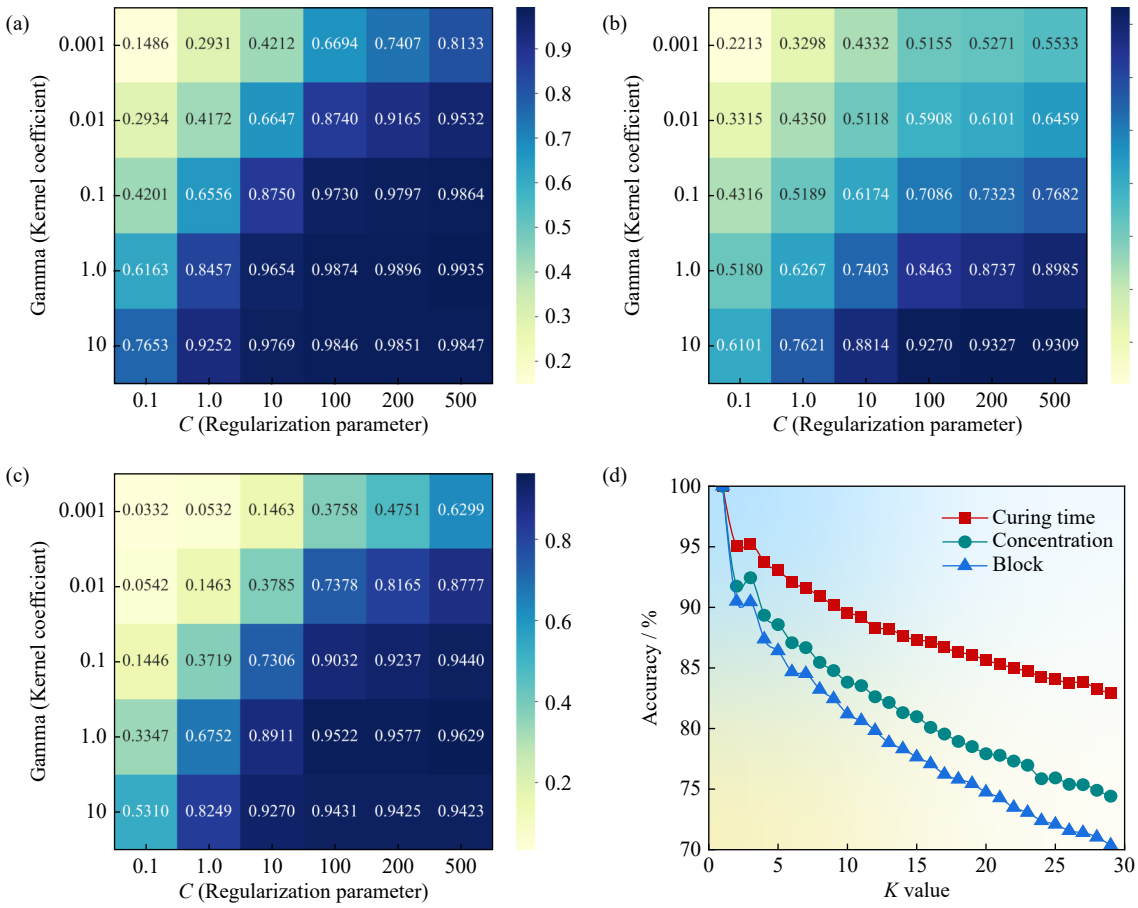

Fig. 8. Hyperparameter optimization process of SVM and KNN algorithm: SVM grid search results for (a) curing time, (b) concentration, and (c) sample block classification; (d) KNN accuracy with varying *K* value.

Table 2. SVM classification results evaluation

Classification category	Data set	Index		
		Accuracy / %	Precision / %	Recall / %
Block	Train	98.49	98.50	98.49
	Test	98.70	98.73	98.70
Concentration	Train	99.31	99.31	99.31
	Test	98.24	98.24	98.24
Curing time	Train	99.83	99.83	99.83
	Test	99.84	99.84	99.84

Table 3. KNN classification results evaluation ($K = 3$)

Classification category	Data set	Index		
		Accuracy / %	Precision / %	Recall / %
Block	Train	90.45	90.67	90.45
	Test	92.64	92.84	92.64
Concentration	Train	92.45	92.47	92.45
	Test	95.03	95.02	95.03
Curing time	Train	95.27	95.35	95.26
	Test	95.26	96.65	96.63

that, whether for the test set or the training set, the SVM algorithm consistently outperforms KNN. This indicates that SVM, through kernel function mapping, transforms the data into a higher-dimensional space where the data may become linearly separable. In contrast, as the dimensionality increases in the KNN algorithm, the data becomes sparser, and the differences in distances between neighbors diminish, which negatively impacts KNN's classification accuracy. Fig. 10 also presents the confusion matrix for the SVM algorithm on the test set, which visualizes the model's performance by showing the correctly and incorrectly classified instances across different categories. This matrix provides a

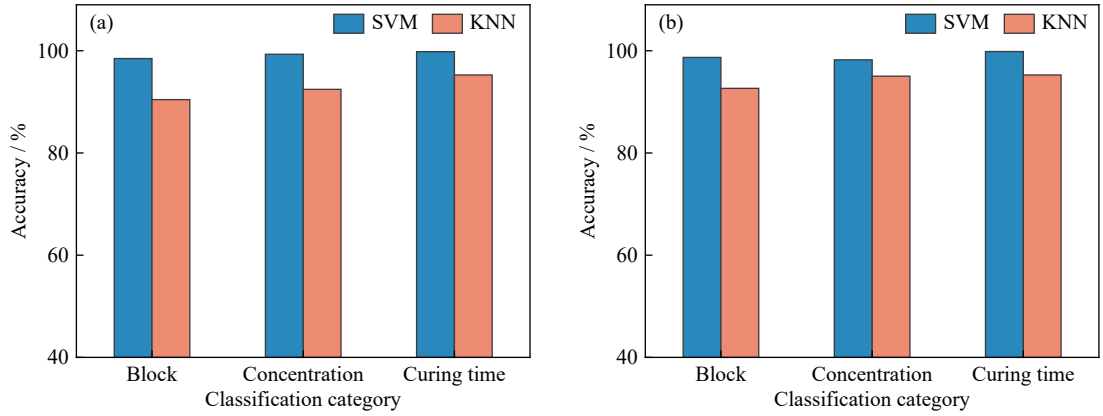

Fig. 9. Comparison of classification accuracy between SVM and KNN: (a) training set; (b) test set.

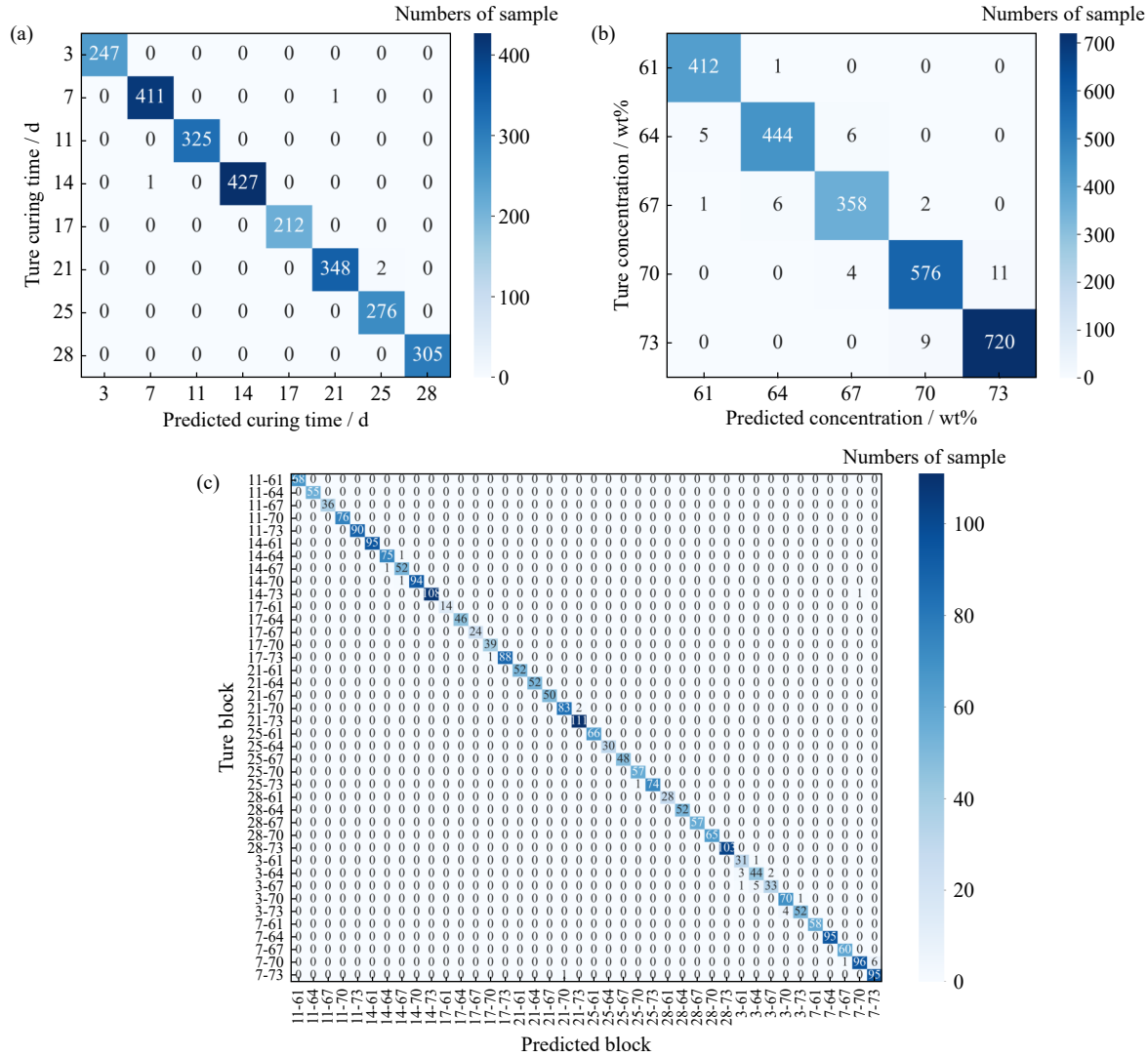

Fig. 10. Confusion matrix for recognition of CPB by SVM algorithm: (a) curing time; (b) concentration; (c) block. Note: the axis label "11-61" denotes a 61wt% concentration specimen cured for 11 d, other labels follow the same format (the first number indicates curing time, the second indicates concentration).

clearer understanding of the specific areas where the SVM algorithm excels or makes errors.

4.4. Shortcomings and future prospects

This manuscript pioneers the introduction of HSI technology into the domain of CPB concentration identification, overcoming the destructive limitations of conventional detection methods. HSI, integrating the complementary advantages of spectral and imaging information, has demonstrated effectiveness in fields such as soil heavy metal concentration inversion and meat quality inspection [56–57], yet its application in backfilling engineering remains uncharted. Focusing on curing CPB concentration identification—distinct from fresh slurry concentration detection—this study fills the research gap in non-destructive CPB concentration testing under stope, providing critical technical support for intelligent backfilling operations. However, this research represents an initial exploration of CPB concentration identification. The generalizability of the established model is inherently constrained by the homogeneity of experimental materials/environment and the limited sample size, rendering it insufficient for direct engineering guidance. For instance, in actual production, when the concentration of CPB fluctuates greatly, the classification model cannot meet the production requirements. Therefore, it is necessary to establish a concentration regression prediction model to achieve this goal. Experimental data is an important foundation for establishing a model, and the concentration gradient should be reduced according to the production conditions. The larger the amount of experimental data, the better the generalization ability of the model. Then, by using different preprocessing algorithms and regression prediction algorithms, the reflectance of the hyperspectral band is taken as the input and the concentration of CPB as the output. Therefore, future work should prioritize constructing a larger-scale dataset incorporating diverse materials and scenarios to enhance model generalizability. Furthermore, developing effective dimensionality reduction techniques to mitigate data redundancy—while maintaining prediction accuracy—represents another promising avenue to improve computational efficiency, warranting further investigation.

5. Conclusions

To address the challenge of undetectable true concentrations in cured CPB, this paper proposes a non-destructive testing technology for CPB concentration based on HSI and machine learning methods. These algorithm models leverage spectral features, particularly the reflectance values at key wavelengths, to make accurate predictions based on the training set. This approach offers a non-destructive and efficient method to monitor and control CPB properties *in situ*. The main conclusions are as follows.

(1) The tailings used in this study exhibit three primary absorption peaks at approximately 1390, 1950, and 2200 nm. The peak at 1390 nm corresponds to the first overtone stretching vibration of the O–H bond in H_2O molecules ($2\nu_1$, $2\nu_3$, and $\nu_1 + \nu_3$). The absorption peak at 1950 nm is associated with the combination of stretching and bending vibrations of the O–H bond.

(2) This study confirms that HSI can effectively capture the spectral characteristics of cured CPB, with its reflectance curves exhibiting significant correlation with concentration. Notably, two distinct absorption peaks are observed near 1407 and 1917 nm. These peaks are influenced by roughness and moisture content, leading to differences in reflectance for different morphologies. Spectral reflectance is particularly sensitive to CPB concentration. As the concentration increases, the overall reflectance of the spectrum gradually increases. However, the area of the absorption peaks at 1407 and 1917 nm also expands, which is inferred to be related to Si–O bond vibrations in C–S–H. A strong linear relationship between CPB concentration and reflectance at 1407 nm was observed.

(3) For classification based on concentration, the highest accuracy of 98.24% is achieved with $C = 200$ and Gamma = 10. When $K = 3$, the model achieves the highest accuracy and stability, with an accuracy of 95.03%. In both the training and test sets, the accuracy of the SVM algorithm surpasses that of KNN.

This study preliminarily demonstrates the feasibility and scientific validity of using HSI for the non-destructive analysis of CPB concentration. However, constrained by limitations such as a finite sample dataset and homogeneous experimental materials, the generalizability of the model remains insufficiently validated, precluding its direct application to industrial production. In the future, HSI can be further utilized to detect indicators such as strength and water content of CPB, thus promoting the intelligent development of CPB.

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Conflict of Interest

Aixiang Wu is the Editor-in-Chief and Qiusong Chen is an editorial board member for this journal. They were not involved in the editorial review or the decision to publish this article. The authors declare that they have no conflicts of interest.

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